

# Introduction to Microeconomics

## Supply & Demand

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Years  
Anniversary

ECON101  
Spring 2026

- Why do male soccer players earn more than female soccer players?
- Why do diamonds cost more than water?
- Why do Picasso's paintings sell for more than other artists?
- Is it the cost of production that determines prices, or is it willingness to pay?
- Alfred Marshall (1890) was the first to clearly explain how both costs and willingness to pay interact to determine market prices.

## Definition: Market

The market for any good or service consists of all (actual or potential) buyers or sellers of that good or service.

## The supply–demand framework

- A *market* is not necessarily a physical place.
  - Mechanism through which buyers and sellers interact.
- The supply–demand model is a *workhorse* of microeconomics.
- Prices and quantities: **endogenous variables**.
- Preferences, technology, income, and other determinants: **exogenous variables** (*ceteris paribus*).
- Focus is on *single markets*, or at most a few interrelated but mostly independent markets.
  - This is the **partial equilibrium** approach.

## The demand function

- Demand is represented by a function:

$$q^D = D(p; \alpha)$$

- It gives the *total quantity* of a good or service that buyers wish to buy.
  - As a function of its own price  $p$ .
  - And of other factors  $\alpha$  (preferences, income, prices of related goods, expectations, etc.).
- The *demand curve* is a **ceteris paribus** analysis.
  - Fix the determinants.
  - Analyze how quantity demanded changes with respect to its own price.

## From demand function to demand curve

- Fix a level of determinants:  $\bar{\alpha}$ .
- Price  $p$ : **exogenous variable**.
- Quantity demanded  $q^D$ : **endogenous variable**.
- The demand curve is the *inverse demand*:

$$p = D^{-1}(q; \bar{\alpha})$$

- It represents the pairs  $(q, p)$  at which buyers are willing to purchase the good.

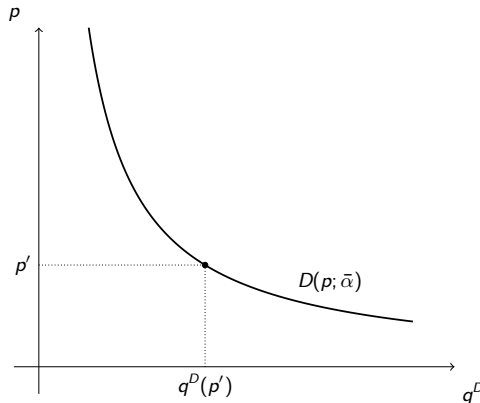
## Law of Demand

- As price increases:
  - the good becomes less affordable,
  - buyers substitute toward cheaper alternatives.
- Quantity demanded decreases.

## Interpreting the demand curve

**Horizontal interpretation:** at price  $p$ , how much quantity is demanded?

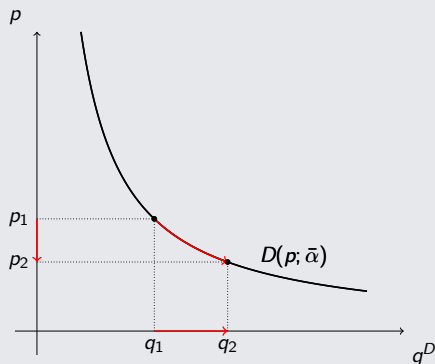
**Vertical interpretation:** at quantity  $q$ , what is the buyer's **willingness to pay**?



# Movements along the demand curve and shifts of the demand schedule

## Changes in quantity demanded

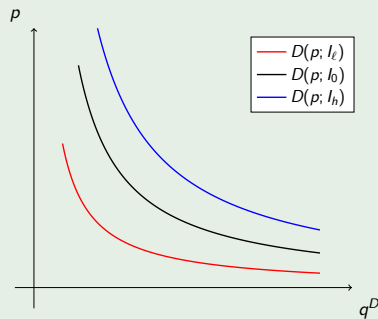
A price change from  $p_1$  to  $p_2$  induces a change in *quantity demanded* from  $q_1$  to  $q_2$ . This is a movement along the demand curve.



## Shifts of Demand

Suppose there is a change in  $\bar{\alpha} \rightarrow \hat{\alpha}$ . This corresponds to a **shift of the demand curve**.

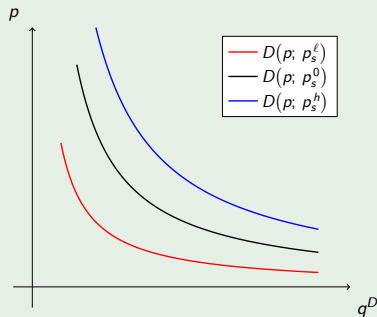
(i) Changes in income: Consider  $0 < I_\ell < I_0 < I_h$ .



# Price of substitutes and complements

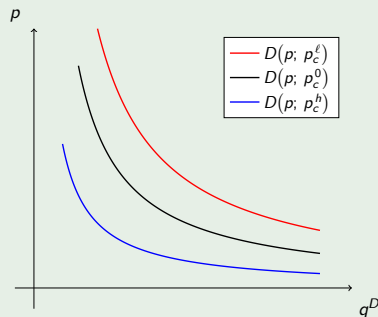
## Shifts of Demand: Price of a Substitute

(ii) Consider the following prices of a substitute good:  $0 < p_s^\ell < p_s^0 < p_s^h$ .



## Shifts of Demand: Price of a Complement

(iii) Consider the following prices of a complement good:  $0 < p_c^\ell < p_c^0 < p_c^h$ .



## The supply function

- Supply is represented by a function:

$$q^S = S(p; \gamma)$$

- It gives the *total quantity* of a good or service that sellers wish to sell.
  - As a function of its own price  $p$ .
  - And of other factors  $\gamma$  (input costs, technology, number of sellers, expectations, etc.).
- The *supply curve* is a **ceteris paribus** analysis.
  - Fix the determinants.
  - Analyze how quantity supplied changes with respect to its own price.

## From supply function to supply curve

- Fix a level of determinants:  $\bar{\gamma}$ .
- Price  $p$ : **exogenous variable**.
- Quantity supplied  $q^S$ : **endogenous variable**.
- The supply curve is the *inverse supply*:

$$p = S^{-1}(q; \bar{\gamma})$$

- It represents the pairs  $(q, p)$  at which sellers are willing to offer the good for sale.

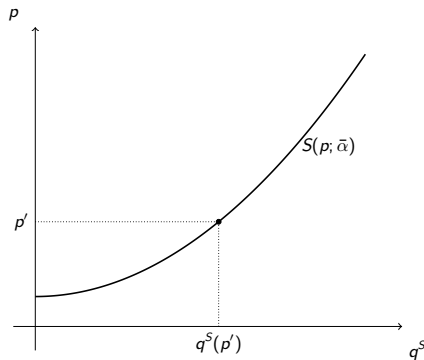
## Law of Supply

- As price increases:
  - producing/selling more becomes worthwhile,
  - marginal (opportunity) cost is increasing.
- Quantity supplied increases.

## Interpreting the supply curve

**Horizontal interpretation:** at price  $p$ , how much quantity is supplied?

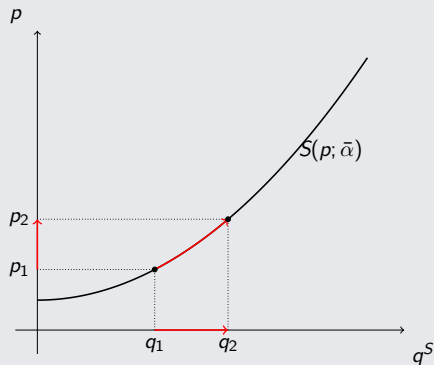
**Vertical interpretation:** at quantity  $q$ , what is the **minimum price** sellers are willing to accept?



# Movements along the supply curve and shifts of the supply schedule

## Changes in quantity supplied

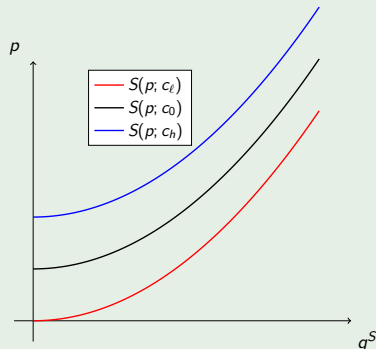
A price change from  $p_1$  to  $p_2$  induces a change in *quantity supplied* from  $q_1$  to  $q_2$ . This is a movement along the supply curve.



## Shifts of Supply

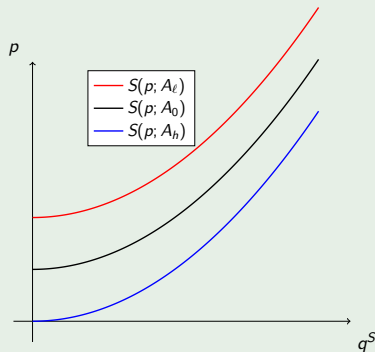
Suppose there is a change in  $\bar{\gamma} \rightarrow \hat{\gamma}$ . This corresponds to a **shift of the supply curve**.

(i) Changes in input costs: Consider  $0 < c_\ell < c_0 < c_h$ .



## Shifts of Supply

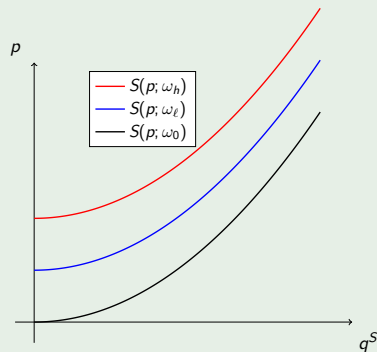
(ii) Technological change: Consider  $A_\ell < A_0 < A_h$ .



## Shifts of Supply

Suppose there is a natural disaster affecting production conditions.

(iii) Natural shock: Consider  $\omega_0 < \omega_\ell < \omega_h$ .



## Consumer Surplus

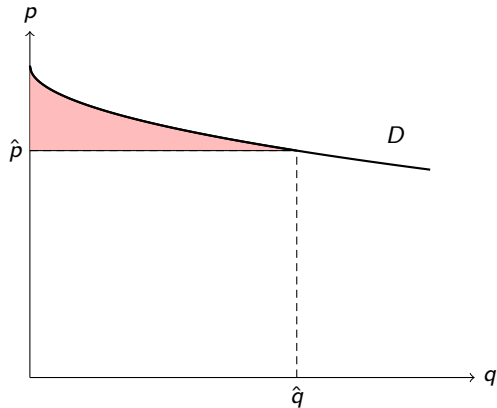
**Consumer Surplus** is the excess of the benefit received from a good over the amount paid for it. We can calculate consumer surplus as the marginal benefit (or value) of a good minus its price, summed over the quantity bought. It is measured by the area under the demand curve and above the price paid, up to the quantity bought.

## Producer Surplus

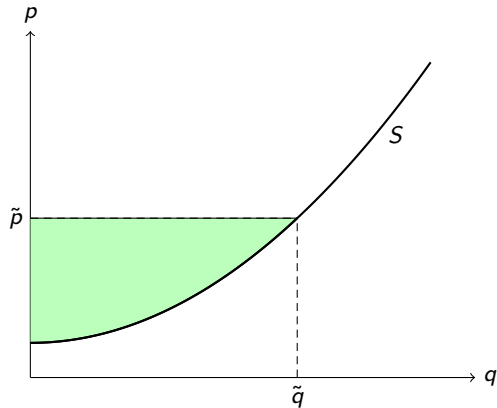
**Producer Surplus** is the excess of the amount received from the sale of a good over the cost of producing it. We calculate it as the price received for a good minus the minimum-supply price (marginal cost), summed over the quantity sold. On a graph, producer surplus is shown by the area below the price received and above the supply curve, summed over the quantity sold.

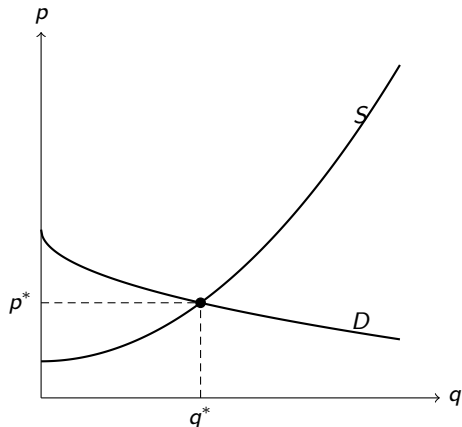
# Consumer and Producer Surplus

## Consumer Surplus



## Producer Surplus





## Market equilibrium

Equilibrium occurs at the price–quantity pair for which both buyers and sellers are satisfied.

For a fixed level of the determinants of demand and supply,  $\alpha$  and  $\gamma$ , a **market equilibrium** is a pair

$$(q^*, p^*)$$

such that

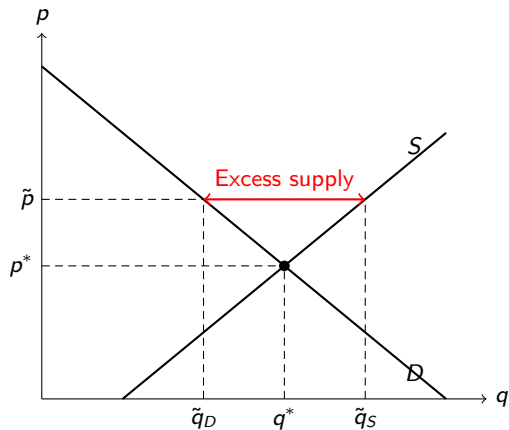
$$q^* = D(p^*; \alpha),$$

$$q^* = S(p^*; \gamma).$$

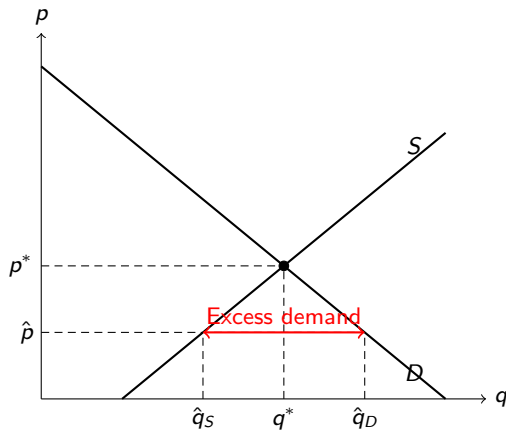
Equilibrium is a *consistency condition*: individual plans are mutually compatible at  $(q^*, p^*)$ .

- Buyers are willing to buy  $q^*$  units at price  $p^*$ .
- Sellers are willing to sell  $q^*$  units at price  $p^*$ .

# Excess Supply and Excess Demand



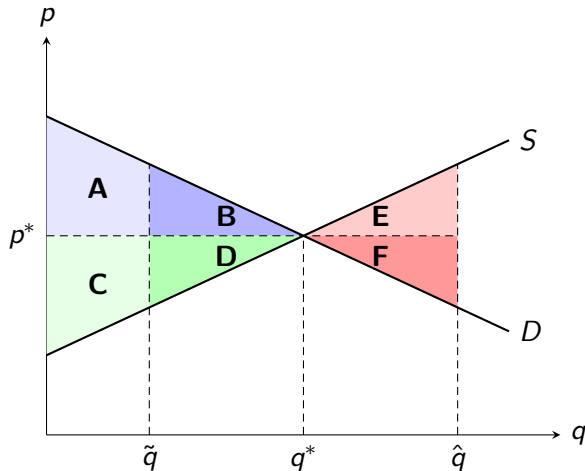
$$\text{Excess Supply}(\tilde{p}) = \tilde{q}_S - \tilde{q}_D$$



$$\text{Excess Demand}(\hat{p}) = \hat{q}_D - \hat{q}_S$$

# Social Surplus

Social surplus is the sum of producer and consumer surplus. It represents the total gains from participating in a market.



## Case 1: Underproduction

Suppose  $\tilde{q}$  is traded at price  $p^*$ :

- $CS = A$
- $PS = C$

## Case 2: Equilibrium

What happens at  $q^*$  and  $p^*$ ?

- $CS = A + B$
- $PS = C + D$

## Case 3: Overproduction

Suppose  $\hat{q}$  is traded at price  $p^*$ :

- $CS = A + B - F$
- $PS = C + D - E$

Note: Trading beyond  $q^*$  creates negative surplus.

## The Equilibrium Principle

A market in equilibrium leaves no unexploited opportunities for individuals, but may not exploit all gains achievable through collective action.

At equilibrium prices:

- Each buyer and seller is doing the best they can, given the prices.
- No individual has an incentive to change their behavior unilaterally.
- Yet, outcomes may still be inefficient from a collective perspective.
  - Specifically, in the presence of externalities or public good environments.

# Linear demand and supply curves

## Linear Demand Function

Consider a linear demand represented by:

$$q^D = \alpha - \beta p$$

where parameters  $\alpha > 0$  and  $\beta > 0$ .

## Deriving Inverse Demand

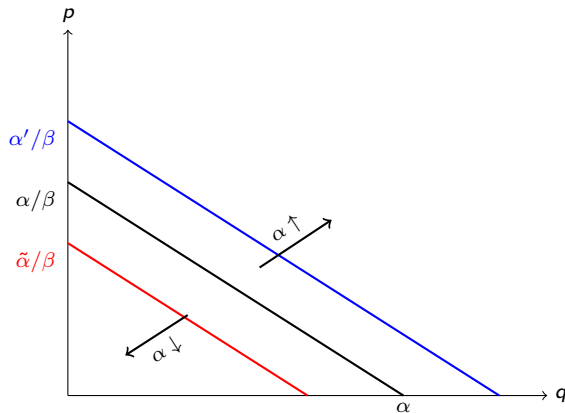
To graph the demand curve, we solve for price ( $p$ ):

- ①  $\beta p = \alpha - q^D$
- ②  $p = \frac{\alpha}{\beta} - \frac{1}{\beta} q^D$

This gives the **vertical intercept** at  $\alpha/\beta$  and the **slope** as  $-1/\beta$ .

## Demand Determinants

$\alpha$  encapsulates all **demand determinants** under consideration (ceteris paribus).



Changes in  $\alpha$ :  $\alpha' > \alpha > \tilde{\alpha}$

# Linear Supply and Shifts

## Linear Supply Function

Consider a linear supply represented by:

$$q^S = \gamma + \delta p$$

where  $\delta > 0$  is the slope and  $-\infty < \gamma < \infty$ .

## Deriving Inverse Supply

To graph the supply curve, we solve for price ( $p$ ):

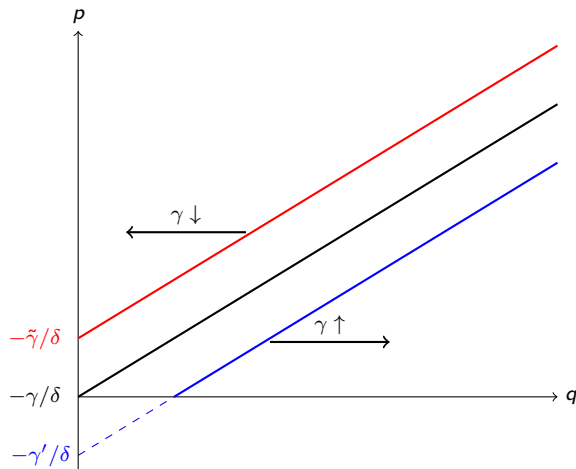
①  $\delta p = q^S - \gamma$

②  $p = -\frac{\gamma}{\delta} + \frac{1}{\delta} q^S$

The **vertical intercept** is  $-\gamma/\delta$  and the **slope** is  $1/\delta$ .

## Supply Determinants

$\gamma$  encapsulates determinants such as technology or input costs.



Changes in  $\gamma$ :  $\gamma' > \gamma > \tilde{\gamma}$

# Market Equilibrium: Linear Case

## The Linear System

Consider the following market model:

$$q^D = \alpha - \beta p \quad (\alpha > 0, \beta > 0)$$

$$q^S = \gamma + \delta p \quad (\delta > 0, \gamma \in \mathbb{R})$$

## Solving for Equilibrium Price ( $p^*$ )

Set  $q^D = q^S$ :

$$\alpha - \beta p^* = \gamma + \delta p^*$$

$$\alpha - \gamma = (\beta + \delta)p^*$$

$$p^* = \frac{\alpha - \gamma}{\beta + \delta}$$

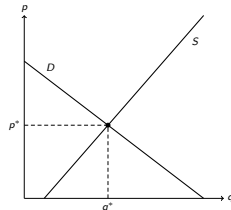
## Solving for Equilibrium Quantity ( $q^*$ )

Substitute  $p^*$  into the demand function:

$$q^* = \alpha - \beta \left( \frac{\alpha - \gamma}{\beta + \delta} \right)$$

$$q^* = \frac{\alpha(\beta + \delta) - \beta(\alpha - \gamma)}{\beta + \delta}$$

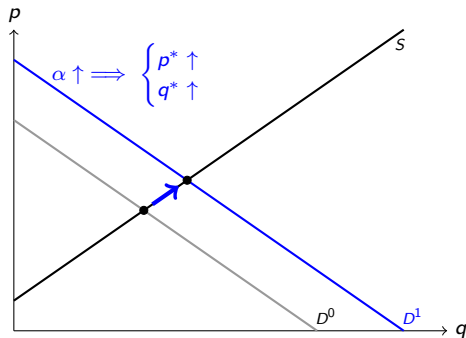
$$q^* = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$



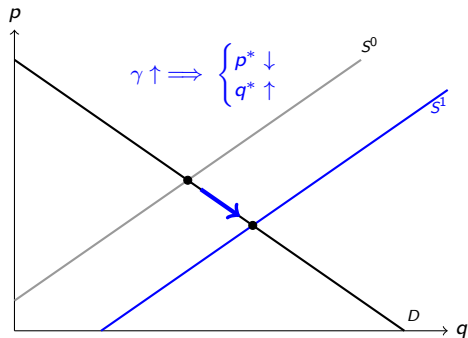
$$p^* = \frac{\alpha - \gamma}{\beta + \delta}$$

$$q^* = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$

Shift of Demand ( $\alpha \uparrow$ )



Shift of Supply ( $\gamma \uparrow$ )

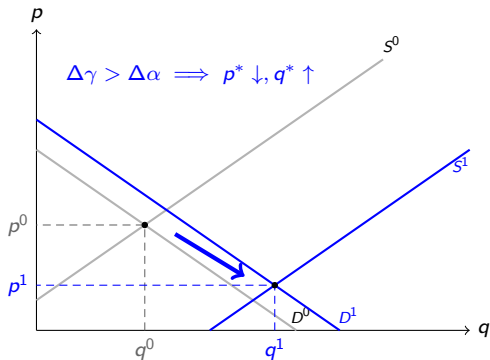


# Simultaneous Shifts: Demand and Supply

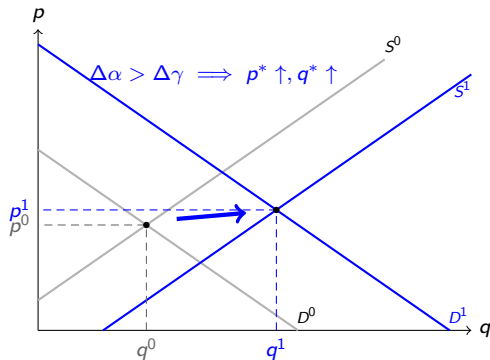
$$p^* = \frac{\alpha - \gamma}{\beta + \delta}$$

$$q^* = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$

**Scenario 1:**  $\alpha \uparrow < \gamma \uparrow$



**Scenario 2:**  $\alpha \uparrow > \gamma \uparrow$



## Homework: Remaining Simultaneous Shifts

$$p^* = \frac{\alpha - \gamma}{\beta + \delta}$$

$$q^* = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$

### Task: Graphical and Comparative Statics Analysis

For each of the following scenarios, draw a diagram illustrating the shifts and determine the sign of  $\Delta p^*$  and  $\Delta q^*$ . Identify which change is **unequivocal** and which is **ambiguous**.

#### 1 Both Curves Decrease:

- Scenario A:  $\alpha \downarrow > \gamma \downarrow$  (Demand shift dominates)
- Scenario B:  $\alpha \downarrow < \gamma \downarrow$  (Supply shift dominates)

#### 2 Opposing Shifts (Case 1):

- $\alpha \uparrow$  and  $\gamma \downarrow$

#### 3 Opposing Shifts (Case 2):

- $\alpha \downarrow$  and  $\gamma \uparrow$

#### 4 The Limit Cases:

- What happens to  $p^*$  if  $\Delta\alpha = \Delta\gamma$ ?
- What happens to  $q^*$  if  $\Delta\alpha = -\frac{\beta}{\delta}\Delta\gamma$ ?

*Hint: Use the boxed formulas above to verify your graphical results algebraically.*

## Aggregate Demand and Supply

Suppose that individual demand curves are given by  $q_i^d(p)$ , for  $i = 1, \dots, I$  consumers; and individual supply curves are given by  $q_k^s(p)$ , for  $k = 1, \dots, K$  producers/firms.

The **aggregate or market demand** is obtained by **horizontal summation** of all the individual demand curves — at price  $p$ , we add up all demanded quantities:

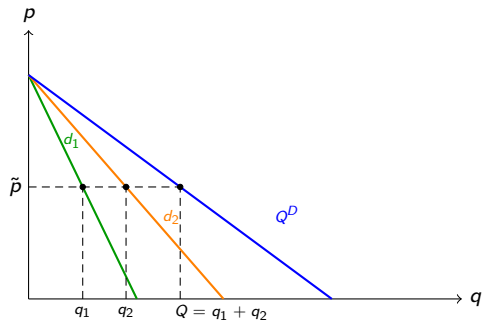
$$Q^D(p) = \sum_{i=1}^I q_i^d(p).$$

Similarly, the **aggregate or market supply** is obtained by **horizontal summation** of all the individual supply curves — at price  $p$ , we add up all supplied quantities:

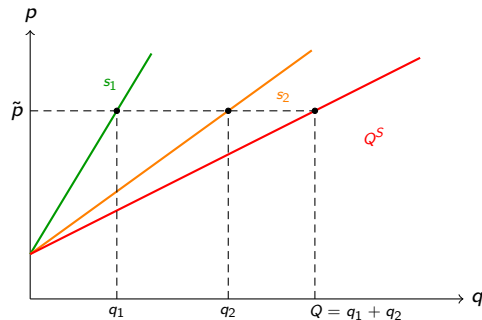
$$Q^S(p) = \sum_{k=1}^K q_k^s(p).$$

# Horizontal Summation

Demand Side:  $Q(p) = \sum q_i(p)$



Supply Side:  $Q(p) = \sum q_k(p)$



*Horizontal Summation: Summing  $q_1$  and  $q_2$  at price  $\tilde{p}$*

## Definition: Effective Domains

To ensure economic meaningfulness, demand and supply must be restricted to regions where quantities are non-negative. We define the sets of feasible price-determinant pairs as:

$$\mathcal{P}_d \times \mathcal{A} = \{(p, \alpha) \in \mathbb{R}_+ \times \mathbb{R} : q^d(p; \alpha) \geq 0\}$$

$$\mathcal{P}_s \times \mathcal{D} = \{(p, \gamma) \in \mathbb{R}_+ \times \mathbb{R} : q^s(p; \gamma) \geq 0\}$$

Thus, the well-defined functions are:

$$q^d : \mathcal{P}_d \times \mathcal{A} \rightarrow \mathbb{R}_+ \quad \text{and} \quad q^s : \mathcal{P}_s \times \mathcal{D} \rightarrow \mathbb{R}_+$$

## Warning: Domain restrictions impact how you aggregate!

### Warning: Piecewise Aggregation

Fix determinantes  $\alpha$  and  $\delta$ , which for clarity, I will suppress from the notation in what follows. The aggregate curves  $Q^D(p)$  and  $Q^S(p)$  are the sum of individual quantities, but each term must be evaluated within its specific effective domain:

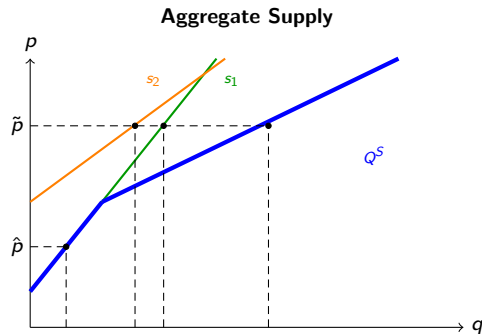
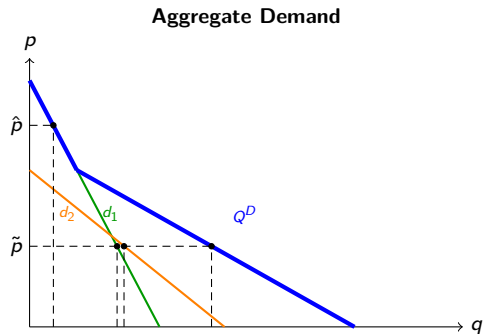
$$Q^D(p) = \sum_{i=1}^I \mathbb{I}_{\mathcal{P}_{d,i}}(p) q_i^d(p) \quad \text{and} \quad Q^S(p) = \sum_{k=1}^K \mathbb{I}_{\mathcal{P}_{s,k}}(p) q_k^s(p)$$

where  $\mathbb{I}_A(\cdot)$  is the indicator function:

$$\mathbb{I}_A(p) = \begin{cases} 1 & \text{if } p \in A, \\ 0 & \text{if } p \notin A. \end{cases}$$

Failure to account for this warning regarding the individual effective domains leads to incorrect horizontal summation of the curves.

# Horizontal Summation: Piecewise Aggregation



# Aggregation: $n$ Identical Consumers

## 1. Effective Price Domain

For a linear individual demand  $q(p) = \alpha - \beta p$ :

- We require  $q(p) \geq 0 \implies \alpha - \beta p \geq 0$ .
- Since  $\alpha > 0$ , the domain for prices is:

$$\mathcal{P}_d = \{p \in \mathbb{R}_+ : p \leq \alpha/\beta\}$$

- This is well-defined provided  $\alpha > 0$ .

## 2. Market Aggregation

With  $n$  identical consumers, the aggregate quantity is:

$$Q^D(p) = n \cdot q(p) = n\alpha - n\beta p$$

## 3. Aggregate Inverse Demand

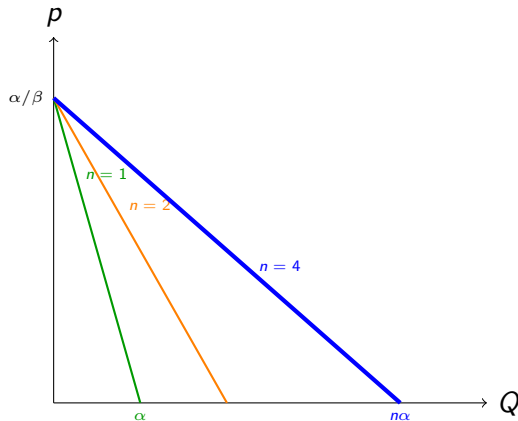
To graph in  $(Q, p)$  space, we invert the function:

$$\textcircled{1} \quad n\beta p = n\alpha - Q^D$$

$$\textcircled{2} \quad p = \frac{n\alpha}{n\beta} - \frac{1}{n\beta} Q^D \implies p = \frac{\alpha}{\beta} - \frac{1}{n\beta} Q^D$$

The vertical intercept remains  $\alpha/\beta$ , while the slope becomes flatter at  $-1/n\beta$ .

## The Impact of $n$ on Market Demand



*As  $n \uparrow$ , the curve rotates outward around the vertical intercept  $a/b$ .*

# Aggregation Example: Different Slopes, Same Intercept

## 1 Individual Demands

- Consumer 1:  $q_1(p) = 10 - p$ .
- Consumer 2:  $q_2(p) = 20 - 2p$ .
- Both share the same vertical intercept: 10.

## 2 Market Aggregation: Summing the quantities horizontally:

$$Q^D(p) = q_1(p) + q_2(p)$$

$$Q^D(p) = (10 - p) + (20 - 2p)$$

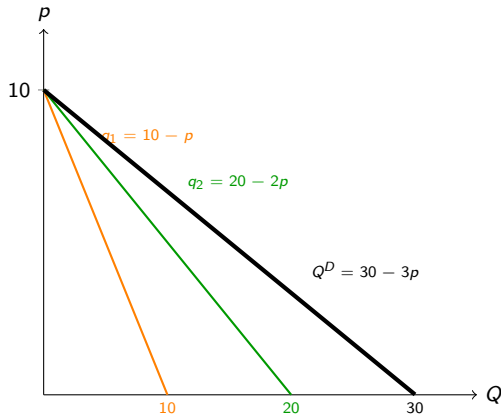
$$Q^D(p) = 30 - 3p$$

## 3 Market Inverse Demand

$$3p = 30 - Q^D \Rightarrow \boxed{p = 10 - \frac{1}{3}Q^D}$$

The slope flattens from  $-1$  and  $-1/2$  to  $-1/3$ .

## Horizontal Summation



# Horizontal Aggregation: Piecewise Demand

## ① Individual Demands and Effective Domains

- **Domestic ( $q_1$ ):**  $q_1(p) = 1430 - 55p$ .
- **Effective Domain:**  $q_1(p) \geq 0 \implies 1430 \geq 55p \implies 0 \leq p \leq 26$ .
- **World ( $q_2$ ):**  $q_2(p) = 1470 - 70p$ .
- **Effective Domain:**  $q_2(p) \geq 0 \implies 1470 \geq 70p \implies 0 \leq p \leq 21$ .

## ② Market Aggregation (Horizontal)

- The aggregate market demand  $Q_D(p)$  sums quantities only where they are non-negative:

$$Q^D(p) = \begin{cases} 1430 - 55p & \text{if } 21 \leq p \leq 26 \\ 2900 - 125p & \text{if } 0 \leq p < 21 \end{cases}$$

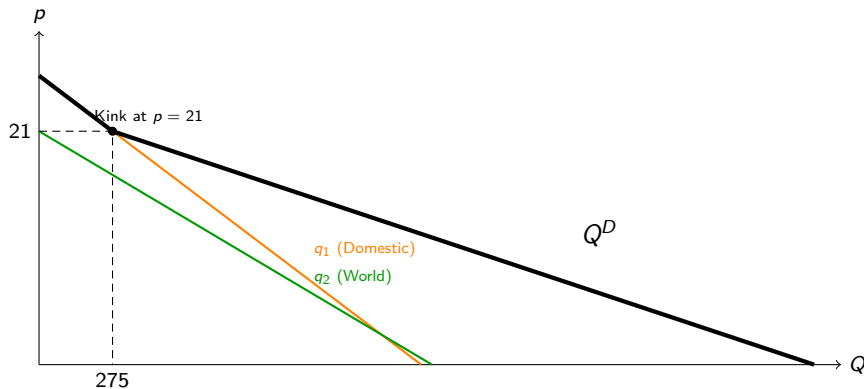
## ③ Piecewise Inverse Demand

- Inverting each segment to find  $p(Q^D)$ :

$$p(Q^D) = \begin{cases} 26 - \frac{1}{55}Q & \text{if } 0 \leq Q^D \leq 275 \\ 23.2 - \frac{1}{125}Q & \text{if } Q^D > 275 \end{cases}$$

- Note: At the kink ( $p = 21$ ),  $Q^D = 1430 - 55(21) = 275$ .

# Graphical Aggregation: The Kinked Demand Curve



*The market curve mimics  $q_1$  until  $p$  drops below the entry price of  $q_2$ .*

# Market Equilibrium

## Market Environment

$$Q^D(p) = \begin{cases} 1430 - 55p & \text{if } p \geq 21 \\ 2900 - 125p & \text{if } p < 21 \end{cases}, \quad Q^{S1}(p) = 5p, \quad Q^{S2}(p) = 100p$$

## Equilibrium for $Q^{S1} = 5p$

**Upper Segment ( $p \geq 21$ ):**

$$5p = 1430 - 55p \Rightarrow 60p = 1430 \Rightarrow p_1^* \approx 23.83.$$

Validation:  $23.83 \geq 21 \Rightarrow$  **VALID**  $\Rightarrow Q_1^* \approx 119.17$ .

**Lower Segment ( $p < 21$ ):**

$$5p = 2900 - 125p \Rightarrow 130p = 2900 \Rightarrow p \approx 22.31.$$

Validation:  $22.31 > 21 \Rightarrow$  **INVALID**.

## Equilibrium for $Q^{S2} = 100p$

**Upper Segment ( $p \geq 21$ ):**

$$100p = 1430 - 55p \Rightarrow 155p = 1430 \Rightarrow p \approx 9.22.$$

Validation:  $9.22 < 21 \Rightarrow$  **INVALID**.

**Lower Segment ( $p < 21$ ):**

$$100p = 2900 - 125p \Rightarrow 225p = 2900 \Rightarrow p_2^* \approx 12.89.$$

Validation:  $12.89 < 21 \Rightarrow$  **VALID**  $\Rightarrow Q_2^* \approx 1289$ .

